Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 5

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: `Python3@Lab` → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: `HELLO123` → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: `hi@#9` → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

In [6]:
```
string = input("Enter a String :")
upperCase = lowerCase = Digits = Special = 0
for i in string1:
    if i.isupper():
        upperCase += 1
    elif i.islower():
        lowerCase += 1
    elif i.isdigit():
        Digits += 1
    else :
        Special += 1

print(f"UpperCase : {upperCase} , LowerCase : {lowerCase} , Digits : {Digits} , Special : {Special}")
```

```
UpperCase : 0 , LowerCase : 2 , Digits : 1 , Special : 2
```

2) Reverse Each Word

- Input: `Python is Easy` → `nohtyP si ysaE`
- Input: `Hello World` → `olleH dlroW`
- Input: `Learn Python Fast` → `nraeL nohtyP tsaF`

In [52]:
```
string = input("Enter a String (enter extra space at last)  :")
temp = 0;
for i in range(0,len(string)):
    if string[i] == " ":
        print(string[i : temp-1 if temp > 0 else None : -1] , end= " ")
        temp = i+1
```

```
nohtyp  si  ysae
```

In [18]:
```
string = input("Enter a String :").split(" ")
for i in string:
    print(i[::-1] , end=" ")
```

```
nohtyp si ysae
```

3) Case Pattern Identifier

- `PYTHON` → Uppercase
- `python` → Lowercase
- `Python Programming` → Title Case

In [53]:
```
string = input("Enter a String :")
if string.isupper():
    print("Uppercase")
elif string.islower():
    print("Lowercase")
elif string.istitle():
    print("Titlecase")
```

```
Uppercase
```

4) First and Last Occurrence Finder

- `programming` , `g` → First:3 Last:10
- `banana` , `a` → First:1 Last:5
- `hello` , `z` → Not found

In [56]:
```
string = input("Enter a String :")
char1 = input("Enter for First and Last Occurrence :")
First = string.find(char1)
Last = string.rfind(char1)
if first == -1:
    print(f"{char1} Not Found")
else:
    print(f"First : {First} Occurrence , Last : {Last} Occurrence")
```

```
First : 3 Occurrence , Last : 10 Occurrence
```

5) Word Frequency Counter (ignore the case)

- `Python is easy and Python is powerful` → python: 2
- `Java is popular but java is verbose` → java: 2
- `C is fast` → c:1

In [57]:
```
string = input("Enter a String :").lower()
substring = input("Enter a String :").lower()
print(f"{substring} : {string.count(substring)}")
```

```
python : 2
```

6) Remove Extra Spaces

- `  Python   is   fun  ` → `Python is fun`
- `  Hello   World` → `Hello World`
- `NoExtraSpace` → `NoExtraSpace`

In [69]:
```
string = input("Enter a String :")
newstring1 = string.strip().split()
newstring2 = " ".join(newstring1)
print(newstring2)
```

```
python is fun
```

7) Prefix and Suffix Removal

- `unhappy.txt` → `happy`
- `pretest.py` → `test`
- `unwanted.doc` → `wanted`

In [71]:
```
string = input("Enter a String :")
prefix = input("Enter prefix to remove :")
suffix = input("Enter suffix to remove :")
print(string.removesuffix(suffix).removeprefix(prefix))
```

```
happy
```

8) Replace Vowels

- `Programming` → `Pr*gr*mm*ng`
- `Education` → `*d*c*t**n`

- `Sky` → `Sky`

In [75]:
```
string = input("Enter a String :")
vowel = 'aeiouAEIOU'
for i in vowel:
    string =string.replace(i ,"*")
print(string)
```

```
pr*gr*mm*ng
```

9) String Compression

- `aaabbcccc` → `a3b2c4`
- `xxxyyyz` → `x3y3z1`
- `abcd` → `a1b1c1d1`

In [88]:
```
string = input("Enter a String :")
temp = 1 ;
res = ""
for i in range(0,len(string)-1):

    if string[i] == string[i+1]:
        temp+=1
    else:
        res += string[i] + str(temp)
        temp = 1
res += string[i] + str(temp)
print(res)
```

```
a3b2c4
```

10) Toggle Case (without using str.swapcase())

- `PyThOn` → `pYtHoN`
- `HELlo` → `helLO`
- `Python` → `pYTHON`

In [78]:
```
string = input("Enter a String :")
res = ""
for i in string:
    if i.isupper():
        res+=i.lower()
    elif i.islower():
        res+=i.upper()
print(res)
```

```
pYtHoN
```

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- `User123` → Valid
- `123User` → Invalid
- `Us@12` → Invalid

In [4]:
```
string = input("Enter a String :")
if len(string) < 6:
    print("invalid")
elif not string.isalnum():
    print("invalid")
elif string[0].isdigit():
    print("invalid")
else:
    print("valid")
```

```
invalid
```

12) Palindrome Checker (ignore the case)

- `Madam` → Palindrome
- `Level` → Palindrome
- `Python` → Not Palindrome

In [80]:
```
string = input("Enter a String :").lower()
if string == string[::-1]:
    print("Palindrome")
else :
    print("Not Palindrome")
```

```
Palindrome
```

13) Longest Word Finder

- `Python programming is interesting` → programming
- `I love coding` → coding
- `Data structures and algorithms` → structures

In [12]:
```
string = input("Enter a Sentence :").split()
print(string)
max1 = string[0]
for i in range(len(string)-1):
    if len(string[i]) < len(string[i+1]):
        max1 = string[i+1]
print(max1)
```

```
['I', 'love', 'coding']
coding
```

14) Case Conversion

- `python programming`
  - Upper: PYTHON PROGRAMMING
  - Lower: python programming
  - Title: Python Programming
  - Capitalize: Python programming

In [16]:
```
string = input("Enter a Word :")
print(f"Upper : {string.upper()}")
print( f"Lower : {string.lower()}" )
print(f"Title : {string.title()}")
print(f"Capitalize : {string.capitalize()}")
```

```
Upper : PYTHON PROGRAMMING
Lower : python programming
Title : Python Programming
Capitalize : Python programming
```

15) Custom Split

- `apple,banana,grape`
- `10|20|30`
- `A-B-C-D`

In [20]:
```
string = input("Enter the word :")
remover = input("Enter the remover :")
newstring = string.split(remover)
print(newstring)
```

```
['10', '20', '30']
```

In [ ]: